light and dry so that they can be carried to long distances by wind.

- In some plants anthers burst violently and release the pollen into the air. Example: *Urtica.*
- Stigmas are comparatively large, protruding, sometimes branched and feathery, adapted to catch the pollen grains. Generally single ovule is present.
- Plant produces flowers before the new leaves appear, so the pollen can be carried without hindrance of leaves.

Pollination in Maize (*Zea mays*): The maize is monoecious and unisexual. The male inflorescence (tassel) is borne terminally and female inflorescence (cob) laterally at lower levels. Maize pollens are large and heavy and cannot be carried by light breeze. However, the mild wind shakes the male inflorescence to release the pollen which falls vertically below. The female inflorescence has long stigma (silk) measuring upto 23 cm in length, which projects beyond leaves. The pollens drop from the tassel is caught by the stigma (Figure 1.15).

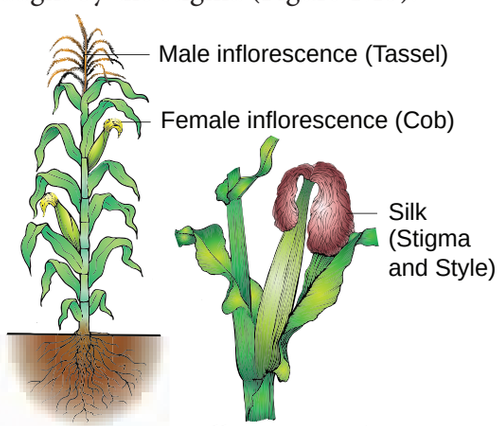


Figure 1.15 Pollination in *Zea mays*

2. Hydrophily: Pollination by water is called hydrophily and the flowers pollinated by water are said to be **hydrophilous** (Example: *Vallisneria, Hydrilla*). Though there are a number of aquatic plants, only in few plants pollination takes place by water. The floral envelop of hydrophilous plants are reduced or absent. In water plants like *Eichhornia* and water lilly pollination takes place through wind or by insects. There are two types of hydrophily, Epihydrophily and Hypohydrophily. In most of the hydrophilous flowers, the pollen grains possesses mucilage covering which protects them from wetting.

a. Epihydrophily: Pollination occurs at the water level. Examples: *Vallisneria spiralis, Elodea.*

Pollination in *Vallisneria spiralis*: It is a dioecious, submerged and rooted hydrophyte. The female plant bears solitary flowers which rise to the surface of water level using a long coiled stalk at the time of pollination. A small cup shaped depression is formed around the female flower on the surface of the water. The male plant produces male flowers which get detached and float on the surface of the water. As soon as a male flowers comes in contact with the female flower and pollination takes place, Stalk of the female flower coils and goes under water where fruits are produced. (Figure 1.16).

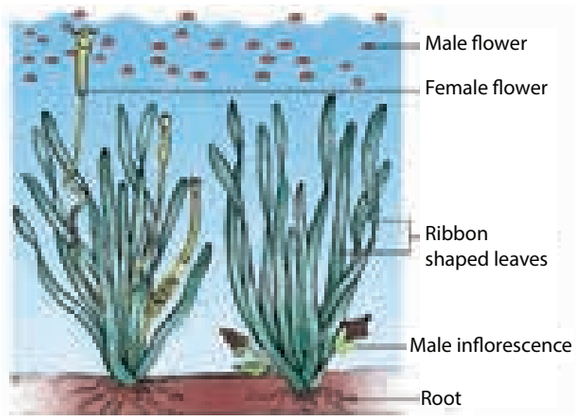


Figure 1.16 Pollination in *Vallisneria*

Activity

Visit to a nearby park and observe the different flowers. Record the adaptations or modifications found in the flowers for different types of pollination.

b. Hypohydrophily: Pollination occurs inside the water. Examples: *Zostera marina and Ceratophyllum.*

Zostera marina is a submerged marine sea grass and pollination takes place under water. The pollen grains are long, needle like. They are shed

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